Studies of temperature change in the mesosphere using OH–emission: Tools for monitoring climatic change.

Cand. Scient. Thesis

by

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May 1998

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Abstract: - Atmospheric temperatures from the mesosphere are deduced from spectrometer measurements of hydroxyl bands night airglow emission. Studying the temperature change in the mesosphere is a key tool for monitoring the climatic change and for prediction of the global warming. The OH (6-2) bands have been observed at a site close to Skibotn in northern Norway (69.5 ^{0}N, 20.3 ^{0}E, geographic) during January-February 1997 using Ebert-Fastie spectrometer. The background theory and instrumental set-up are described in detail, followed by a critique of sources of error. This part of the thesis provides complete background information to enable future users of the instrument to obtain temperature data quickly and easily. Necessary software is included in the appendices. An average temperature of 190.3 ± 7 K for the January campaign is obtained from a selected data set. Variations in rotational temperature at about 87 km on time scales from tens of minutes to days show similar pattern as MSISE90 at the same altitude and latitude region. The semidiurnal tide with average peak to peak amplitude of ~ 8 K is obtained. The average horizontal tidal velocity amplitude deduced from the amplitude of the semidiurnal temperature variation is 17 m s^{-1}. The effects of gravity wave propagation in the mesosphere on the rotational temperature are also discussed. Finally, strategies combining OH and incoherent scatter observations to investigate ion chemistry and support turbulence measurements are presented.

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